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REVIEW

Aldosterone and the Mineralocorticoid Receptor in Atrial Fibrillation

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ABSTRACT: Atrial fibrillation represents the most prevalent cardiac arrhythmia and is associated with substantial morbidity, including an increased risk for stroke and heart failure. The pathophysiology of atrial fibrillation involves electrical and structural remodeling of the atria, often referred to as atrial myopathy, that together increase the risk for arrhythmias. However, a specific approach to target the proarrhythmic substrate of atrial fibrillation is still lacking. Aldosterone and the mineralocorticoid receptor are well-known drivers of cardiac remodeling, and recent clinical and experimental studies indicate that they play a critical role in the pathogenesis of atrial fibrillation. Elevated aldosterone levels, for example, in primary aldosteronism, are associated with a higher risk for atrial fibrillation. Mineralocorticoid receptor antagonists reduce the onset of atrial fibrillation across various patient populations, including patients with hypertension, heart failure, chronic kidney disease, or undergoing cardiac surgery. In patients with preexisting atrial fibrillation, mineralocorticoid receptor antagonists may decrease atrial fibrillation recurrence when added to antiarrhythmic therapies. Experimental studies provide a direct link between aldosterone and atrial remodeling and arrhythmia. Mineralocorticoid receptor activation modulates several key cellular processes involved in atrial inflammation, fibrosis, and arrhythmogenesis, including fibroblast activation, cardiomyocyte dysfunction, and ion channel activity. Here, we review what is currently known about the role of aldosterone and the mineralocorticoid receptor in atrial fibrillation, summarize the mechanistic basis as supported by experimental studies, and discuss the potential of mineralocorticoid receptor antagonists in the prevention and treatment of atrial fibrillation.

Key Words: atrial fibrillation ■ fibroblasts ■ fibrosis ■ heart failure ■ myocytes, cardiac

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Atrial fibrillation (AF) is the most prevalent cardiac arrhythmia, characterized by rapid and disorganized atrial activity. AF affects ≈1% of the general population, with prevalence rising to nearly 10% among the elderly.[1] From 2010 to 2019, the global prevalence of AF surged from 33.5 million to 59 million individuals.[2] AF increases the risk of serious medical complications, including stroke and heart failure (HF), that influence patient outcomes.[2] Genome-wide association studies have identified an array of genetic mutations that are associated with AF, many of them affecting genes coding for ion channels, structural or contractile proteins, or transcription factors.[3] However, in many cases AF may be acquired since higher age and comorbidities such as hypertension, HF, or coronary artery disease increase the risk of developing AF.[3] Current treatment of AF is largely based on 3 main pillars: anticoagulation therapies for the prevention of stroke, antiarrhythmic or heart rate–lowering drugs for rhythm or rate control, and catheter ablation.[4] In addition, it is recommended to treat risk factors and comorbidities to reduce the recurrence of AF.[4] However, a specific approach to target the proarrhythmic substrate of AF is still lacking.[5]

The pathophysiology of AF involves electrical, mechanical, and structural remodeling of the atria, which together increase the risk for arrhythmias. Dysregulation of sodium, potassium, and calcium channels, as well as intracellular calcium handling proteins in atrial cardiomyocytes, reduces the action potential (AP) duration with the consequence of shortening the effective refractory period, increasing the risk of reentry, or causing spontaneous atrial ectopic activity. Spontaneous ectopic activity often occurs in the pulmonary veins.[6] Pulmonary vein ostia contain muscular tissue, and it has been proposed that the chaotic arrangement of muscle fibers found here promotes electrical reentry.[7] On the other hand, structural abnormalities, such as interstitial fibrosis, adipocyte infiltration, or disturbed

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Nonstandard Abbreviations and Acronyms

α-SMA	α-smooth muscle actin
11β-HSD2	11-β-hydroxysteroid dehydrogenase type 2
AF	atrial fibrillation
AF-CHF	Atrial Fibrillation and Congestive Heart Failure
AP	action potential
AT1	angiotensin II type 1 receptor
COL1A	collagen 1A
COL3A	collagen 3A
CREBH	cAMP-responsive element–binding protein H
CTGF	connective tissue growth factor
ECM	extracellular matrix
ERK	extracellular signal–regulated kinase
HF	heart failure
HFpEF	HF with preserved ejection fraction
I_{Kur}	ultra-rapid rectifier potassium current
JNK	c-Jun N-terminal kinase
$K_{v1.5}$	voltage-gated potassium channel 1.5
LOX	lysyl oxidase
MAPK	mitogen-activated protein kinase
MR	mineralocorticoid receptor
MRA	mineralocorticoid receptor antagonists
PKA	protein kinase A
Smad	mothers against decapentaplegic homolog
TGFβ1	transforming growth factor β1

cell-cell contacts, form the substrate for reentry, leading to AF.[8] This process is often referred to as atrial myopathy.[3] Increased atrial stretch due to volume overload, reactive oxygen species (ROS), and inflammation are important drivers of atrial myopathy.[8] Importantly, these processes are self-perpetuating, giving rise to the concept that AF begets AF, in which sustained arrhythmia accelerates atrial remodeling and disease progression. In this review article, we discuss preclinical and clinical studies that indicate that aldosterone and the MR (mineralocorticoid receptor) take a central role in AF pathogenesis and may serve as new therapeutic target.

BIOLOGY OF ALDOSTERONE-MR SIGNALING

The MR is a member of the nuclear receptor family that is bound by aldosterone or cortisol.[9] After ligand binding, MR shuttles from the cytosol to the nucleus, where it acts as a transcription factor by directly interacting with hormone response elements in the DNA, thereby regulating gene expression.[9] Aldosterone is a steroid hormone synthesized in the zona glomerulosa of the adrenal cortex in response to angiotensin II or high serum potassium.[10] Aldosterone and cortisol bind with similar affinity to the MR.[11] The enzyme 11β-HSD2 (11-β-hydroxysteroid dehydrogenase type 2), converting cortisol into inactive cortisone, enables aldosterone-MR binding.[12] In addition, MR activation may occur independent of aldosterone. Rac1 (Ras-related C3 botulinum toxin substrate 1), a member of the Rho family GTPases, directly activates MR signaling in vitro and in vivo independently of circulating aldosterone concentrations.[13,14] Besides genomic effects, aldosterone induces rapid, nongenomic effects altering intracellular signaling, potentially MR-independent.[15] Although nongenomic effects of aldosterone are only partly MR-independent, as several pharmacological and genetic studies show, some nongenomic effects of aldosterone still require the classical MR.[16–18] In the classical model, aldosterone via MR in epithelial cells of the kidney and the colon enhances sodium reabsorption by increasing activity and expression of apical epithelial sodium channels and basolateral Na^+/K^+-ATPase pumps.[19] More recently, MR functions have been described across a broad spectrum of tissues and organs, including the kidney, the lung, the liver, and the cardiovascular system.[20,21] In the heart and the vasculature, MR activation causes tissue inflammation, fibrosis, and remodeling.[22–25] Experimental studies using MR gene deletion in mice revealed MR-dependent effects in cardiomyocytes, fibroblasts, smooth muscle cells, endothelial cells, and various immune and inflammatory cells.[22] For instance, cardiomyocyte MR deletion protected from cardiac injury after myocardial infarction,[26] chronic pressure overload,[27] or deoxycorticosterone/salt-induced hypertension.[28]

THERAPEUTIC TARGETING OF ALDOSTERONE AND THE MR

MR antagonists (MRA) have been well-established in the treatment of HF and arterial hypertension for decades.[29] Clinical trials have demonstrated that MRA reduced mortality in chronic HF with reduced ejection fraction,[30,31] leading to a class IA recommendation in current treatment guidelines.[32,33] Furthermore, spironolactone is a potent blood pressure–lowering drug and recommended in patients with resistant arterial hypertension.[34] Recently, the FIGARO-DKD and FINEARTS-HF phase 3 clinical trials have demonstrated that the MRA finerenone effectively decreases the risk of HF events in patients with chronic kidney disease and type 2 diabetes at risk[35] or with established HF with preserved ejection fraction.[36]

First and second generation MRA, spironolactone and eplerenone, are steroidal-backbone competitive

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antagonists similar to aldosterone. Third-generation MRAs, including finerenone, esaxerenone, and balcinrenone, have nonsteroidal structures and bind MR with high affinity and specificity.[37,38] Across clinical trials, MRA use is associated with about 2-fold increased risk for hyperkalemia.[39] Recently, newly developed inhibitors of aldosterone synthase (CYP11B2) have emerged as an alternative to MRA that may offer advantages in conditions like primary hyperaldosteronism but leave MR unprotected from cortisol or ligand-independent Rac activation.[40]

The accumulating evidence that aldosterone and MR drive tissue inflammation and fibrosis has raised the question of whether MRA could provide benefits in indications beyond HF.[21] Recent experimental and clinical studies have investigated aldosterone and MR roles across diverse indications, including chronic kidney disease,[41,42] heart valve disease,[43] pulmonary hypertension,[44] skin disease,[45] or retinopathy.[46] Here, we will summarize evidence from experimental and clinical studies on aldosterone and MR in AF and myopathy and discuss the cellular and molecular processes involved.

IMPACT OF MRAS ON ATRIAL REMODELING AND ARRHYTHMIAS IN CLINICAL TRIALS

A series of clinical studies explored the potential of MRAs to attenuate atrial remodeling and electrophysiological abnormalities and reduce AF burden (Table 1).[47–65] Multiple meta-analyses consistently demonstrate that MRAs reduce AF risk by 23% to 32% across diverse patient cohorts.[66–68] Several studies have examined MRA effects on AF in patients with HF with variable results depending on disease severity and phenotype. In HF with reduced ejection fraction populations, spironolactone improved atrial remodeling and reduced AF episodes.[47–49] However, a post hoc analysis of the AF-CHF study (Atrial Fibrillation and Congestive Heart Failure) revealed that MRA use in patients with AF with severe left ventricular dysfunction (left ventricular ejection fraction ≤35%) and New York Heart Association class II to IV symptoms was associated with increased cardiovascular mortality over 6-year follow-up, mainly driven by sudden cardiac death.[52] The authors propose that MRA use, mostly spironolactone, could have contributed to hyperkalemia, leading to an increased risk for arrhythmia; however, due to the retrospective nature of the study and the lack of respective data, this remains speculative.[52] In HF with preserved ejection fraction (HFpEF), the TOPCAT trial showed no significant effect of spironolactone on AF occurrence.[57] Conversely, the RACE 3 trial demonstrated that MRA-based upstream therapy in persistent patients with AF with early HF (both HFpEF and HF with reduced ejection fraction) supported sinus rhythm maintenance over 1 year.[59] Beyond spironolactone and eplerenone, recent evidence indicates the benefits of finerenone, a nonsteroidal MRA, in AF. For example, a pooled analysis of the FIDELIO-DKD, FIGARO-DKD, and FINEARTS-HF trials revealed that finerenone significantly reduced new-onset AF/atrial flutter in patients at high risk and with established HFpEF.[62] However, when specifically examining patients with HFpEF based on prespecified analysis of the FINEARTS-HF trial, finerenone yielded a nonsignificant 23% relative reduction in new-onset AF/atrial flutter.[69] Meta-analyses directly comparing patient subgroups show that patients with left ventricular dysfunction experience significantly greater AF risk reduction with MRA therapy compared with those with preserved ventricular function,[70] with effect size augmented by longer treatment duration.[70] Taken together, MRAs seem to improve atrial remodeling and reduce AF burden in mild to moderate HF with reduced ejection fraction, whereas they provide little or no benefit for AF outcomes in HFpEF. When examining broader cardiovascular outcomes, MRAs reduced cardiovascular death or HF hospitalization similarly in patients with HF and without HF, regardless of baseline AF status.[71] However, MRAs show superior efficacy in preventing recurrent AF episodes (risk reduction 27%–58%) compared with new-onset AF (risk reduction 16% to 50%).[67,68,70]

The impact of MRAs on AF has been assessed in various patient collectives beyond HF. In patients with essential hypertension or after myocardial infarction, MRA improved atrial electrophysiological abnormalities along with atrial fibrosis markers.[50,56] Spironolactone significantly reduced atrial remodeling and improved atrial function independent of blood pressure in resistant hypertension.[63] Furthermore, MRA treatment in primary aldosteronism mitigated increased AF risk comparably to surgical adrenalectomy.[60] In cardiac surgery patients, MRAs are being investigated as a potential AF preventive approach. Retrospective studies showed reduced postoperative AF after cardiac surgery with eplerenone and spironolactone.[55,65] In prospective randomized trials, results are inconsistent. Spironolactone reduced AF incidence after coronary bypass surgery in a cohort of patients with mildly reduced EF,[64] whereas in an earlier study in patients with normal ejection fraction undergoing cardiac surgery, it had no effect.[53] A recent study showed that preoperative MRA use (median 63 days) reduced both early postoperative AF (19.8% versus 33.2% within 30 days) and long-term paroxysmal or chronic AF up to 6 years after cardiac surgery.[72] The ALDOCURE trial (URL: https://www.clinicaltrials.gov; Unique identifier: NCT03551548) is expected to evaluate MR blockade efficacy in preventing postoperative AF in 1500 patients with HF undergoing elective cardiac surgery.[73]

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Table 1. Summary of Available Clinical Trials Showing the Benefits of MRA on Atrial Remodeling and Arrhythmia

Study details	MRA treatment strategy	Main effects of MRAs	Summary	Reference
Study design: randomized, double-blind, placebo-controlled trial. Patient population: patients with HF (n=116). Study aim: to assess if SPL improves magnesium homeostasis and reduces arrhythmias in patients with HF	Treatment groups: (1) SPL group (SPL 20 mg/d; n=58), (2) placebo control group (n=58). Follow-up: 6 mo	SPL caused slower HR and fewer atrial ectopic beats (24 h monitoring). SPL caused fewer AF compared with placebo group (22% vs 41%; ARR 19% [95% CI, 2%–36%]; *P*=0.046).	SPL reduces atrial arrhythmias in patients with HF.	Gao et al[47]
Study design: prospective, randomized, open-label clinical trial Study population: paroxysmal patients with AF (n=164) Study aim: to evaluate the antiarrhythmic effects of SPL compared with ACE inhibitors in recurrent patients with AF	Treatment groups: (1) Group A (SPL 25 mg/d+β-blocker+enalapril (n=41). (2) Group B: SPL 25 mg/d+β-blocker (n=41). 3) Group C: enalapril+β-blocker (n=41). Group D: β-blocker alone (control; n=41) Follow-up: 12 mo	SPL reduced AF episodes. SPL was more effective than enalapril or β-blocker alone in reducing AF episodes. Addition of enalapril did not provide added benefit vs SPL+β-blocker alone.	SPL with combination of β-blockers prevents AF episodes in patients with maintained LV function and a history of refractory paroxysmal AF	SPIR-AF study, Dabrowski et al[48]
Study design: a prospective study Patient population: patients with HF (n=21) Study aim: to examine the effects of SPL on atrial conduction and remodeling in patients with HF	Treatment groups: (1) SPL group (SPL 25 mg/d; n=11), (2) control group (n=10). Follow-up: 12 mo	SPL reduced LA diameter. SPL shortened P-wave duration and P-wave amplitude on ECG.	SPL improves atrial conduction and remodeling in patients with HF	Kimura et al[49]
Study design: randomized follow-up study Patient population: first patients with ST-segment–elevation AMI (n=110), successfully revascularized with percutaneous coronary intervention Study aim: to evaluate the effect of spironolactone on atrial remodeling in patients with preserved LV functions after AMI.	Treatment groups: (1) Conventional therapy group (n=55), (2) SPL group (SPL 25 mg/d+conventional therapy group; n=55) Follow-up: 6 mo	SPL increased in LA emptying fraction-SPL prevented the prolongation of the time from P-wave onset to the end of atrial contraction wave in the LA, interatrial septum, and RA	SPL provides mild benefits on atrial remodeling and electromechanical properties in patients with AMI with preserved LV functions.	Kayrak et al[50]
Study design: retrospective cohort study Study population: patients with AF and structural heart disease (n=83) Study aim: to determine if SPL has beneficial effects on AF	Treatment groups: (1) SPL group (SPL+conventional therapy; n=23), (2) Control (conventional therapy; n=60) Follow-up: ≥3 mo	SPL caused fewer AF-related hospitalizations or direct current cardioversions. SPL lowered the risk of AF events for 79% with an adjusted relative risk of 0.21 ([95% CI, 0.064–0.69]; *P*=0.009).	SPL reduces AF-related hospitalizations or cardioversion for AF	Williams et al[51]
Stud design: a post hoc analysis of data from the AF-CHF trial Patients: patients with AF with LV systolic dysfunction (LVEF≤35%), and HF symptoms (NYHA class II–IV; n=1376). Study aim: to assess the impact of MRAs on cardiovascular mortality in patients with AF and HF enrolled in the AF-CHF trial.	Groups: (1) MRA group, (2) control group (therapy without MRAs). follow-up: up to 6 y	MRAs increased in total mortality for 1.4-fold (HR, 1.4; [95% CI, 1.1–1.8]; *P*=0.005). MRAs increased cardiovascular mortality by 1.4-fold (HR, 1.4; [95% CI, 1.1–1.9]; *P*=0.009). MRAs increased risk of sudden cardiac death by 2-fold (HR, 2.0 [95% CI, 1.3–3.0]; *P*=0.001).	MRAs were independently associated with adverse outcomes.	Post hoc analysis of AF-CHF trial, O'Meara et al[52]
Study design: randomized, double-blind, placebo-controlled clinical trial Patient population: patients undergoing cardiac surgery (n=445) Study aim: to investigate the risks and benefits of an ACE inhibitor or MRA in patients undergoing cardiac surgery	Treatment groups: (1) placebo group (n=147), (2) ramipril group (n=151), 3) SPL group (n=147)	SPL did not decrease the prevalence of postoperative AF (25.9% in SPL group vs 27.2% in Pl group vs 27.8% in ramipril group)	Neither ACE nor MR blockade decreased the primary outcome of postoperative AF	Pretorius et al[53]
Study design: a retrospective, observational study Patient population: patients with long-standing persistent AF who underwent RFCA (n=161). Study aim: to evaluate the effects of EPL on clinical outcomes after RFCA in patients with long-standing persistent AF.	Treatment groups: (1) EPL group (n=55), (2) control group (non-EPL group; n=106). Follow-up duration: 24 mo	Over 24 mo, 60% of EPL-treated patients stayed AF-free compared with 40% in the noneplerenone group (*P*=0.011)	EPL significantly improved the maintenance of sinus rhythm after RFCA in patients with long-standing persistent AF	Ito et al[54]

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Table 1. Continued

Study details	MRA treatment strategy	Main effects of MRAs	Summary	Reference
Study design: a retrospective, single-center, nonrandomized cohort study. Patient population: patients with HFrEF (n=332) without a history of AF. Study aim: to evaluate the impact of the MRAs in preventing new-onset postoperative AF in patients undergoing cardiac surgery (CABG or valve surgery).	Treatment groups: (1) MRA group (n=132). (2) Non-MRAs group (n=200) Follow-up: up to 30 d	MRAs lowered POAF for 30.3% compared with the non-MRAs treatment (45%; $P<0.01$, χ^2 test). MRA treatment and LA diameter were predictors of POAF development. Both EPL and SPL reduced POAF incidence. MRAs attenuated LA enlargement and increased P-wave duration.	MRAs significantly reduced the incidence of POAF when added to standard HF therapy in patients undergoing on-pump cardiac surgery	Simopoulos et al[55]
Study design: a randomized, open-label controlled trial Patient population: patients with HTN (n=35). Aim of the study: comparing the effects of EPL vs amlodipine for AF prevention based on P-wave morphology in patients with HTN	Treatment groups: (1) EPL group (n=16), (2) amlodipine group (n=19). Follow-up: 12 mo	Compared with amlodipine EPL maintained P-wave duration, decreased P-wave dispersion, and decreased the levels of the collagen biomarkers PINP and PIIIP	EPL prevented PV wave morphology changes and fibrosis marker alterations	Kawasaki et al[56]
Study design: post hoc analysis of the TOPCAT trial Patient population: patients with HFpEF enrolled in North and South America (n=1765) Aim of the study: to assess the relationship between AF and outcomes in patients with HFpEF, evaluate if AF modified the treatment response to SPL, and determine if SPL influenced postrandomization AF	Treatment groups: SPL group vs placebo group Follow-up: 2.9 y	SPL did not influence the occurrence of postrandomization AF (HR, 0.98 [95% CI, 0.68–1.42]; $P=0.92$). There was no effect of SPL on the risk of stroke in patients with AF at enrollment (HR, 1.17 [95% CI, 0.64–2.16]; $P=0.60$)	SPL did not show any significant effects on preventing new-onset AF or modifying AF-related outcomes in patients with HFpEF.	The TOPCAT Trial, Cikes et al[57]
Study design: a retrospective cohort analysis based on data from the ORBIT-AF registry Patient population: nonpermanent patients with AF (n=7012)Aim of the study: to evaluate the impact of MRA use on AF progression and cardiovascular outcomes in a contemporary AF population	Treatment groups: (1) Baseline MRA users; (2) New MRA users Follow-up: 2.3±0.8 y	MRA did not reduce AF progression (HR, 1.18 [95% CI 0.88–1.58]; $P=0.27$). MRA lowered a risk of stroke, TIA, or systemic embolism (HR, 0.17 [95% CI, 0.02–1.23]; $P=0.08$).	MRA use in patients with AF, primarily driven by HF and HTN, did not significantly slow AF progression but showed a trend towards reducing stroke risk.	ORBIT-AF registry, Fudim et al[58]
Study design: the RACE 3 trial was a prospective, multicenter, randomized, open-label, blinded end point trial Patients: patients with HFpEF or HFrEF with persistent AF (n=245). Study aim: to test if MRAs added to conventional therapy improve sinus rhythm maintenance in patients with persistent AF and early HF	Treatment groups: (1) MRA group (n=119) and (2) control group (n=126) Follow-up: 1 y	MRAs-based upstream therapy increased 1-y sinus rhythm maintenance to 75% vs 63% with conventional therapy (OR, 1.77, $P=0.042$)	In RACE 3 trial, adding SPL and EPL to targeted therapy improved 1-y sinus rhythm maintenance	RACE 3 trial, Rienstra et al [9]
Study design: retrospective cohort study Patient population: patients with PA treated with MRAs (n=195), patients with PA treated with surgical adrenalectomy (n=201), age-matched patients with HTN (n=40 092) Aim of the study: to investigate whether MRA therapy or surgical adrenalectomy in patients with PA influences the risk for incident AF	Treatment groups: (1) MRA group (patients with PA with PRA <1 ng/mL per hour after MRA treatment (insufficient MR blockade; n=130), patients with PA with PRA ≥1 ng/mL per hour after MRA treatment (sufficient MR blockade; n=65). (2) patients with PA who underwent surgical adrenalectomy (n=201). (3) patients with HTN (n=40 092; no MRA treatment) Follow-up: 8 y	Patients with ineffective MR blockade (PRA <1 ng/mL per hour) carry a 2.5-fold higher AF risk vs essential patients with HTN (adjusted HR, 2.55 [95% CI, 1.75–3.71]). Patients with effective MR blockade (PRA ≥1 ng/mL per hour) or surgical adrenalectomy have similar AF risk as essential patients with HTN (adjusted HR, 1.03 [95% CI, 0.54–2.00] and adjusted HR, 0.75 [95% CI, 0.41–1.36], respectively).	In patients with PA, treatment with MRAs that sufficiently block the MR (PRA ≥1 ng/mL per hour) or surgical adrenalectomy mitigates the increased risk for incident AF. However, treatment with MRAs that do not adequately block the MR (PRA <1 ng/mL per hour) is associated with a significantly higher risk for incident AF compared with patients with HTN	Hundemer et al[60]
Study design: a single-center randomized controlled study Patient population: patients with AF (n=203). Study aim: to study the effects of RFCA combined with SPL for the treatment of AF	Treatment groups: (1) SPL group (RFCA+SPL 20 mg/d; n=102), (2) Control group (RFCA alone; n=101). Follow-up: 12 mo	SPL decreased Ang II, ALD, NT-proBNP, and LA size compared with baseline and the ablation at 3 mo. SPL lowered AF recurrence at 3 (18.6% vs 30.7%; $P=0.046$) and 12 mo (14.7% vs 26.7%; $P=0.039$) compared with ablation.SPL increased cumulative 12-month AF-free survival (85.3% vs 73.3%; $P=0.033$).	In patients undergoing RFA, spironolactone lowers early and late AF recurrence possibly due to decreased aldo and Ang II levels	Wang et al[61]

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Table 1. Continued

Study details	MRA treatment strategy	Main effects of MRAs	Summary	Reference
Study design: secondary analysis of the pooled data from FIDELIO-DKD, FIGARO-DKD, and FINEARTS-HF phase III randomized, double-blind, placebo-controlled trials Patient population: patients with CKD and T2D or with HFpEF (n=14 581) Study aim: to examine the effect of finerenone on new-onset AF or AFF and cardiorenal effects by history of AFF	Treatment groups: finerenone vs placebo (1:1 randomization). 10 or 20 mg finerenone once daily Follow-up: 2.9 y	New-onset AFF occurred in 286 (3.9%) participants receiving finerenone and 345 (4.7%) assigned to placebo (HR, 0.83 [95% CI, 0.71–0.97]; *P*=0.019).	In patients with CKD and T2D or with HFpEF, treatment with the nonsteroidal MRA finerenone significantly reduced the risk of new-onset AFF compared with placebo over a median 2.9-year follow-up period.	FINE-HEART pooled analysis, Pabon et al[62]
Study design: prospective clinical study Patient population: patients with resistant HTN (n=36) Study aim: to determine the blood pressure–independent effects of SPL on LA remodeling and function in resistant patients with HTN	Treatment groups: single SPL treatment group. SPL initiated at 25 mg/d, increased to 50 mg/d after 4 wkFollow-up: 6 mo	LA volumes decreased.LA function improved (total LA emptying fraction increased, active LA emptying fraction increased, LA expansion index increased, active LA longitudinal strain increased)	In this 6-mo study of SPL in patients with resistant HPN, treatment was associated with significant reductions in LA size and improvements in LA function independent of blood pressure changes.	Girard et al[63]
Study design: randomized, single-blind, placebo-controlled trial Patient population: CABG patients (n=130) Study aim: to investigate the effect of SPL on the incidence of POAF in patients undergoing CABG surgery.	Treatment groups: (1) Spironolactone group (50 mg/d; n=65), (2) placebo control group (n=65). Follow-up: 2 wk postoperatively.	The SPL group had a lower incidence of POAF (7.7%) compared with the placebo group (20%; OR, 0.33, *P*=0.042).	SPL administration reduces POAF incidence in CABG candidates compared with placebo	Farzaneh et al[64]
Study design: retrospective, propensity-matched cohort study (1:3) of 19 042 adult cardiac surgery patients at Mayo Clinic (1993–2023). Patient population: adults (18–75) undergoing primary sternotomy for CABG, valve surgery, or myectomy, excluding those with prior AF, previous cardiac surgery, or emergent needs. Study aim: to assess if preoperative MRA reduces POAF and long-term AF postsurgery, and to explore cellular mechanisms using single-nucleus RNA sequencing in atrial tissue.	Treatment groups: preoperative MRA group (n=298, propensity-matched). Non-MRA group (n=894, propensity-matched). Median duration of preoperative MRA ≈ 63 d Follow-up: up to 6 y for chronic AF development.	Preoperative MRA use significantly reduced new-onset POAF during the index hospitalization or within 30 d (19.8% vs 31.8%, *P*<0.001).At 6 y, preoperative MRA use lowered the cumulative incidence of paroxysmal/chronic AF, which was 9.2% in the MRA group compared with 18.9% in the non-MRA diuretic group (HR, 0.45; *P*<0.001)	Preoperative MRA use reduces POAF and chronic AF after cardiac surgery.	Danesh et al[65]

ACE, angiotensin-converting enzyme; AF, atrial fibrillation; AF-CHF, Atrial Fibrillation and Congestive Heart Failure; AFF, atrial flutter; ALDO, osmotic minipump delivered aldosterone infusion; AMI, acute myocardial infarction; Ang II, angiotensin II; CABG, coronary artery bypass grafting; CKD, chronic kidney disease; EPL, eplerenone; HF, heart failure; HF, heart failure; HFpEF, heart failure with preserved ejection fraction; HFrEF, heart failure with reduced ejection fraction; HR, hazard ratio; HTN, hypertension; LA, left atrium; LV, left ventricle; LVEF, left ventricular ejection fraction; MRA, mineralocorticoid receptor antagonists; NT-proBNP, N-terminal pro-B-type natriuretic peptide; NYHA, New York Heart Association; OR, odds ratio; PA, primary aldosteronism; POAF, postoperative atrial fibrillation; RA, right atrium; RFCA, radiofrequency catheter ablation; RHTN, resistant hypertension; SPL, spironolactone; and T2D, type 2 diabetes.

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In patients with paroxysmal AF, spironolactone combined with β-blockers proved superior to enalapril or β-blockers alone in reducing AF episodes.[48] Eplerenone maintained sinus rhythm postablation in long-standing persistent AF,[54] whereas spironolactone combined with ablation decreased AF recurrence and improved 12-month AF-free survival.[61] In AF with structural heart disease, spironolactone reduced AF-related hospitalizations by 79%.[51] However, ORBIT-AF registry analysis found no significant AF progression slowing, although it suggested reduced stroke risk.[58]

Collectively, evidence supports MRA efficacy in AF prevention and management across HF, hypertension, primary aldosteronism, and perioperative settings (Table 1). Meta-analysis studies consistently demonstrate MRAs reduce incident or recurrent AF by ≈30% to 45%,[67,70,74] yet these meta-analysis results are predominantly driven by trials with patients with HF in which AF was a secondary end point. Notably, when restricted to placebo-controlled RCTs, the AF benefit lost significance in the meta-analysis,[67] as many included studies of these meta-analyses were based on post hoc analysis, retrospective or prospective observational studies, and randomized controlled studies without placebo. Currently, there are no large-scale, definitive randomized controlled trials focusing exclusively on MRAs for AF prevention and management as the primary outcome.

REGULATION OF ALDOSTERONE AND MR IN AF

Several studies assessed the correlation of aldosterone levels and/ or MR expression in the heart with the risk for AF (Table 2).[75–84] One the one hand, patients with increased aldosterone levels are at higher risk of developing AF. For example, in a 5.5-year follow-up of ARIC observational cohort study, including 4547 participants,

Table 2. A Summary of Available Studies Evaluating Circulating Aldosterone Levels in Atrial Arrhythmia and Remodeling in Various Disease Conditions

Patient population	Main findings	Reference
Chronic persistent patients with AF before electrical cardioversion (n=22)	Aldosterone levels decreased 48 h after successful cardioversion. Aldosterone/renin ratio decreased after cardioversion.	Goette et al[75]
Patients in SR with previous ECG-documented atrial fibrillation (n=158)	Patients with AF at the follow-up visit (2.6 y) had higher aldosterone levels than patients in SR	Dixen et al[76]
Persistent nonvalvular AF (n=43) referred for electrical cardioversion	In patients maintaining SR at 30 d after successful cardioversion, aldosterone decreased. A drop in aldosterone >13.2 pg/mL at 24 h predicted maintained SR at 30 d with 87% sensitivity and 64% specificity	Wozakowska-Kaplon et al[77]
Patients with persistent AF scheduled for elective cardioversion (n=60), patients with permanent AF (n=19), and healthy individuals as a control group (n=19)	Persistent patients with AF had a higher aldosterone level compared with healthy controls. After successful cardioversion and restoration of SR in persistent patients with AF, the aldosterone level decreased significantly at 30 d and 180 d compared with baseline levels before cardioversion. The decrease in aldosterone levels after cardioversion was accompanied by improved atrial remodeling.	Soeby-Land et al[78]
Patients undergoing CABG surgery (n=137) were assessed to determine if preoperative plasma aldosterone levels predict postoperative AF	Patients with postoperative AF were significantly older and had higher preoperative plasma aldosterone levels compared with those without postoperative AF. Older age and higher aldosterone levels were associated with increased postoperative AF. Higher aldosterone levels were associated with a worse LA strain rate	Alexandre et al[79]
Patients with HTN (n=148). The groups: group I (n=75): PAC ≤138 pg/mL and group II (n=73): PAC >138 pg/mL	LA conduit function was decreased in the higher aldosterone group (group II) compared with the lower aldosterone group (group I). Aldosterone was positively correlated with LAVI, E/e′ ratio, and negatively correlated with LAPEF, LA strain (LA S-S, LA S-E), and LA strain rate (LA SR-S, LA SR-E).	Zhang et al[80]
Enrolled patients (n=233) divided into 3 groups: SR group (n=84), paroxysmal AF group (n=76), persistent AF group (n=73)	Aldosterone levels were higher in the persistent AF group compared with the paroxysmal AF and SR groups. Uric acid and LA diameter were independent predictors of aldosterone levels.	Wang et al[81]
Community-based cohort study of 4547 people without HF, who underwent echocardiography	Renin suppression (≤0.5 ng/mL per hour), a marker of MR activation, was present in 11.6% of participants and was associated with greater LAVI. Higher aldosterone levels were associated with greater LAVI. The highest tertile of aldosterone levels was associated with an incident AF over a median follow-up of 5.5 y, compared with the lowest tertile. In participants with renin suppression, higher aldosterone levels were associated with lower lateral LV-E′ velocity.	Brown et al[82]
patients with AF undergoing RFCA (n=153). Groups: (1) recurrence group (n=35, 22.88%): patients who experienced AF recurrence after RFCA, and (2) nonrecurrence group (n=118, 77.12%): patients who did not experience AF recurrence after RFCA	The AF recurrence group had higher preoperative aldosterone levels compared with the nonrecurrence group. Higher preoperative aldosterone levels were associated with increased risk of AF recurrence after RFCA. Patients with preoperative aldosterone levels above 243.61 pg/mL had a higher rate of AF recurrence (37%) compared with those with aldosterone ≤243.61 pg/mL (11%) during follow-up.	Ruan et al[83]
MESA study participants underwent cardiac MRI, and aldosterone levels were measured 7–8 y prior (n=615)	Higher aldosterone associated with greater LAVI Higher aldosterone associated with reduced LA function—lower strain, strain rate, emptying fraction	MESA study, Varadarajan et al[84]

AF indicates atrial fibrillation; CABG, coronary artery bypass grafting; HF, heart failure; HTN, hypertension; LA S-E, left atrial systolic strain during early filling phase; LA SR-E, left atrial strain rate during early filling phase; LA SR-S, left atrial strain rate during systole; LA S-S, left atrial systolic strain during systole; LA, left atrium; LAPEF, left atrial passive emptying fraction; LAVI, left atrial volume index; LV, left ventricle; MESA, Multi-Ethnic Study of Atherosclerosis; MR, mineralocorticoid receptor; MRI, magnetic resonance imaging; PAC, plasma aldosterone concentration; PRA, plasma renin activity; RFCA, radiofrequency catheter ablation; and SR, sinus rhythm.

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higher aldosterone levels were associated with a greater risk of AF.[82] Similarly, the MESA study with 615 participants linked higher aldosterone levels to several signs of left atrial remodeling and dysfunction.[84] High preoperative aldosterone levels predicted higher AF risk after cardiac surgery.[85] Patients with primary hyperaldosteronism have a higher AF rate compared with patients with essential hypertension.[60,86] A meta-analysis found that AF was 3.5-fold more common in hypertensive patients with primary aldosteronism versus essential hypertension, independent of blood pressure.[87] In turn, sufficient pharmacological MR blockade or surgical adrenalectomy reduced AF risk,[60,88] suggesting a causal link between increased aldosterone levels and AF risk. However, it is important to note that AF risk may remain elevated up to 3 years after initiation of treatment when compared with patients with essential hypertension.[89]

On the other hand, patients with AF show higher aldosterone levels compared with patients without AF. Among patients with arterial hypertension and nonvalvular AF, 42% had pathologically increased aldosterone levels.[90] Patients with persistent AF showed higher circulating aldosterone levels than those with paroxysmal AF and sinus rhythm[81] and left atrial remodeling parameters were related to aldosterone levels in patients with AF.[81] In line with this, patients with higher

aldosterone levels were at greater risk for recurrent AF after radiofrequency catheter ablation.[83] Beyond systemic aldosterone increase, a local cardiac renin-angiotensin-aldosterone system may contribute to the overall aldosterone effect. Patients with AF demonstrate upregulated MR[91–93] and 11β-HSD2[92,94] in atrial tissue, associated with increased profibrotic marker expression and atrial tissue fibrosis.[93,94] Although CYP11B2 mRNA was detected in atrial tissue,[93] expression was low, and thus it remains controversial whether there is relevant aldosterone synthesis in cardiac tissue.[95] In summary, these results show that elevated aldosterone levels in the bloodstream are associated with atrial remodeling and arrhythmia (Table 2) and suggest that they reciprocally influence each other.

ALDOSTERONE INFUSION CAUSES ATRIAL REMODELING AND ARRHYTHMIA IN EXPERIMENTAL MODELS

To clarify whether increased aldosterone levels are causal for atrial remodeling and arrhythmia development, different animal models have been used (Table 3).[96–103] Aldosterone infusion together with a high-salt diet and unilateral nephrectomy is an established model to induce arterial hypertension and cardiac remodeling in rodents. Uninephrectomized rats developing atrial fibrosis after receiving aldosterone infusion and 1% NaCl drinking water for 6 weeks was first described in 1997[96] and confirmed several times later in rats[96,98,99] and mice.[100,101] Similarly, aorto-caval shunt combined with aldosterone infusion in rats caused atrial remodeling and increased AF susceptibility.[99] Of note, aldosterone/salt[103] or aldosterone alone[97,102] without nephrectomy is sufficient to induce atrial remodeling and arrhythmia. The aldosterone doses used in these experiments ranged from 40 to 160 μg/kg per day in rats[96–99] and from 120 to 500 μg/kg per day in mice,[101–103] but whether a dose-effect relation exists remains uncertain. Taken together, studies used living animals,[98–100] isolated hearts,[97,100,103] or isolated atrial cardiomyocytes[101] to characterize electrical remodeling and arrhythmia susceptibility in response to aldosterone infusion (Table 3). Furthermore, many of these studies demonstrated that aldosterone infusion is associated with atrial cardiomyocyte hypertrophy[97,103] and fibrosis.[96,97,100,103] However, it is important to note that all of the in vivo models described above lead to more or less pronounced fluid retention, which may cause atrial stretch, an important risk factor for AF. Along that line, patients with hyperaldosteronism show signs of cardiac volume overload that is reversible with spironolactone treatment.[104] It appears, therefore, plausible that hemodynamic alterations in addition to the direct effects of aldosterone contribute to the increased risk for atrial arrhythmia.

IMPACT OF MRAS ON ATRIAL REMODELING AND ARRHYTHMIA IN EXPERIMENTAL STUDIES

Along the lines of the clinical trials described above, several studies have demonstrated that MR inhibition prevents or reverses atrial remodeling and arrhythmia in experimental models in various species and across different cardiac dysfunction models, such as myocardial infarction, rapid atrial pacing, hypertension, or chronic hypoxia (Table 4).[98,105–112] In a rat model of AF caused by chronic intermittent hypoxia, treatment with eplerenone minimized atrial enlargement, attenuated atrial fibrosis, and lowered AF inducibility.[112] In addition, spironolactone blocked atrial electrical and structural remodeling in rats receiving aldosterone infusion.[98] In the rapid atrial pacing model of AF utilizing implanted pacemakers in sheep, eplerenone treatment attenuated atrial remodeling, including atrial dilation, cellular hypertrophy, and interstitial fibrosis during AF progression. Although this intervention delayed the progression to persistent AF, it could not prevent it.[111] In a rabbit rapid atrial pacing model of atrial arrhythmia, eplerenone reduced L-type calcium current in isolated atrial cardiomyocytes.[106] Collectively, MR inhibition using eplerenone or spironolactone consistently reduced atrial remodeling and arrhythmia susceptibility across various animal models by diminishing atrial dilatation, fibrosis, and electrical remodeling (Table 4).

CELL TYPE–SPECIFIC MR FUNCTIONS IN ATRIAL REMODELING

As outlined above, aldosterone and MR exert effects on multiple organ systems, including the heart. In particular conditions with altered hemodynamics, for example, HF, may cause mechanical stress to the atria and thereby indirectly increase the risk for AF. To dissect MR-mediated from indirect effects and to explore the cell type–specific role of MR in cardiac remodeling, several mouse models with selective MR deletion have been developed and tested in various conditions.[22] For example, MR deletion from either macrophages, endothelial cells, or cardiomyocytes attenuated left ventricular inflammation and remodeling in response to deoxycorticosterone/salt treatment[28,113–115]; however, none of these studies reported atrial phenotypes. In the following section, we will discuss what is known about the effects of aldosterone and MR on cardiac cells based on in vitro and in vivo studies.

MR SIGNALING IN ATRIAL CARDIAC FIBROBLASTS

Fibrosis is a key risk factor for the development of arrhythmia as the ECM (extracellular matrix) serves as an insulator, separating cardiomyocyte bundles, and thereby

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Table 3. Summary of Available Studies Evaluating the Aldosterone Infusion as a Model of Atrial Remodeling and Arrhythmias

Model details*	Key changes related to atrial remodeling and arrhythmia in the studied animal models	Reference
Animal: male Sprague-Dawley rats Model: rat ALDO+1%NaCl+UNx model ALDO infusion rate: 60 µg/kg per day Duration: 6 wk	Atrial tissue remodeling (↑RA and LA fibrosis) Atrial molecular remodeling (↑ACE and bradykinin receptor binding)	Sun et al[96]
Animal: Male Sprague-Dawley rats Model: rat ALDO model ALDO infusion rate: 120 µg/kg per day Duration: 8 wk	Atrial arrhythmia (↑susceptibility to AF) Atrial electrical remodeling (↑atrial conduction times and ↑conduction heterogeneity) Atrial tissue remodeling (↑fibrosis, ↑cardiomyocyte size)	Reil et al[97]
Animal: male Wistar rats Model: rat ALDO model ALDO infusion rate: 40 µg/kg per day Duration: 4 wk	Atrial tissue remodeling (↑fibrosis) Atrial arrhythmia (↑AF induction) Atrial molecular remodeling (↑Kir2.1 and Kv1.5)	Lammers et al[98]
Animal: male Wistar rats Model: rat ALDO+ACS model ALDO infusion rate: 160 µg/kg per day Duration: 4 wk	Atrial remodeling (↑LA major axis, ↑atrial weight) Atrial arrhythmia (↑AF induction)	Cao et al[99]
Animals: MD1 knock-out and WT mice Model: mouse ALDO+1%NaCl+UNx model ALDO infusion rate: 144 µg/kg per day Duration: 4 wk	Atrial remodeling (↑LA area) Atrial arrhythmia (↑AF induction, ↑AF duration) Atrial electrical remodeling (↑QTc) Atrial tissue remodeling (↑fibrosis) Diastolic dysfunction (↑Tau, ↑dp/dt min) Atrial molecular remodeling (↑Collagen I, collagen II, CTGF)	Shuai et al[100]
Animals: WT (Trpm4$^{+/+}$) or KO (Trpm4$^{-/-}$) mice (C57BL/6J background) Model: mouse ALDO+1%NaCl+UNx model ALDO infusion rate: 120 µg/kg per day Duration: 4 wk	Atrial remodeling (↑LA diameter) in both WT and TRPM4-KO mice Atrial electrical remodeling (↓APD, ↑EADs, and DADs in isolated atrial cardiomyocytes) in WT mice and (↑APD, ↓EADs, and DADs in isolated atrial cardiomyocytes) from TRPM4-KO mice.	Simard et al[101]
Animals: male and female db/db and WT mice. Model: mouse db/db+ALDO model ALDO infusion rate: 288 µg/kg per day Duration: 4 wk	Atrial remodeling (↑LA area) Diastolic dysfunction (↑E/A, ↑E/E´)	Hegyi et al[102]
Animals: male and female WT mice. Model: mouse ALDO+1%NaCl model ALDO infusion rate: 500 µg/kg per day Duration: 2 wk	Atrial remodeling (↑LA area) Atrial arrhythmia (↑atrial ectopy and irregular activity without P waves) Atrial tissue remodeling (↑cardiomyocyte size, ↑fibrosis) Diastolic dysfunction (↓E/A) Atrial molecular remodeling (altered cardiomyocyte, fibroblasts, and T cells and their cell-cell communication)	Meral et al[103]

ACE, angiotensin-converting enzyme; ACS, aorto-venocaval shunt; AF, Atrial FIBRILLATION; ALDO, osmotic minipump delivered aldosterone infusion; APD, action potential duration; CTGF, connective tissue growth factor; DADs, delayed afterdepolarizations; E/A, ratio of early to late ventricular filling velocities; E/E´, ratio of early mitral inflow velocity to early diastolic mitral annular velocity; EADs, early afterdepolarizations; Kir2.1, inwardly rectifying potassium channel; Kv1.5, voltage-gated potassium channel 1.5; KO, knockout; LA, Left atrium; RA, right atrium; Trpm4, transient receptor potential cation channel subfamily M member 4; UNx, unilateral nephrectomy; WT, wild type.

*For consistency across studies in this table, the aldosterone infusion units were converted to micrograms per kilogram per day (µg/kg per day), with original units available in the cited papers for exact details.

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promotes electrical reentry.[116] As outlined above, several studies demonstrated that MR activation in vivo causes atrial fibrosis.[96,97,100,103] Single-nucleus RNA sequencing from the mouse left atrium indicated increased cell-cell communication between cardiomyocytes, T cells, and fibroblasts after aldosterone/salt treatment.[103] Atrial fibroblasts are the main source of ECM in the heart, and pioneering work already in the 1990s showed that aldosterone dose-dependently increases collagen production by fibroblasts in vitro.[117] Later studies confirmed that and demonstrated that aldosterone stimulation of atrial cardiac fibroblasts increases expression of profibrotic mediators such as CTGF (connective tissue growth factor), LOX (lysyl oxidase), fibronectin, α-SMA (α-smooth muscle actin), or microRNA-21 via MR signaling.[94] We recently found that MR deletion from fibroblasts prevents interstitial fibrosis, diastolic dysfunction, and atrial dilatation in mice exposed to aldosterone/salt,[118] which is in contrast to an earlier study that found no significant effect of MR deletion on interstitial fibrosis after left ventricular pressure overload.[27] This indicates that the role of fibroblast MR depends on the disease context and may be influenced by other biochemical signals as well as hemodynamic factors. Interestingly, MRA treatment prevented the upregulation of CTGF in fibroblasts not only after aldosterone treatment but also after stimulation with TGFβ1 (transforming growth factor β1).[94] In mice with transgenic activation of the TGFβ signaling pathway, eplerenone attenuated atrial fibrosis and macrophage infiltration,[110] suggesting that MR may influence

Table 4. Summary of Available In Vivo Studies Assessing the Effects of MRA in Various Animal Models of Atrial Arrhythmia and Remodeling

Animals and models	MRA application details	The key observed effects	Reference
Male Wistar rats. LAD-occlusion MI (4 mo) LAD-occlusion for 3 mo and 1 mo of drug treatment (4 mo)	Dosages: SPL (10 mg/kg per day), lisinopril (1 mg/kg per day), atenolol (1 mg/kg per day). Drugs were given alone or in combinations: SPL+lisinopril, lisinopril+atenolol, or SPL+lisinopril+atenolol Duration: 1 mo, starting at 3 mo after LAD-occlusion	↓Atrial tissue remodeling (↓fibrosis) ↓Atrial arrhythmia (↓atrial premature complexes, ↓P-wave duration)	Milliez et al[105]
Adult female New Zealand white rabbits Rapid atrial pacing at 600 bpm for 24 h (1 d) and 120 h (5 d)	Dosage: EPL (5 mg/kg), orally twice daily Duration: started 2 d before rapid atrial pacing for 5 d (total 7 d) or started 6 d before rapid atrial pacing for 1 d (total 7 d).	↓Atrial myocyte ion currents (↓$I_{Ca,L}$)	Laszlo et al[106]
Mongrel dogs. Model: rapid atrial pacing model for AF. Duration: 6 wk (after 1-week recovery, 6 wk of atrial pacing at 400 bpm with implanted pacing leads).	SPL (20 mg/kg per day), oral administration. Duration: started 1 wk before atrial pacing and continued during the 6 wk of pacing.	↑Atrial function (↑ LAEF and LAAEF). ↓Atrial remodeling (↓LAVmax, LAVmin, LAAVmax, LAAVmin). ↓Atrial arrhythmia (↓AF inducibility and duration) ↓Atrial molecular remodeling (↓caspase-3, bax, calpain I, and MMP-9 expressions, ↑Bcl-2, calpastatin, and TIMP-1 expression)	Zhao et al[107]
Male DS rats Model: high-salt diet–induced HTN Duration: 7–20 wk	EPL (200 mg/kg per day), oral (mixed in chow) Duration: from 13 wk to 20 wk of age (7 wk)	↓ Atrial arrhythmia (↓AF inducibility) ↓Atrial tissue remodeling (↓atrial cardiomyocyte size, ↓fibrosis) ↓Atrial molecular remodeling (↓CCL2, ↓OPN, ↓TNF-α expressions, ↓inflammatory cell infiltration, ↓HNE staining, ↓Rac1 GTPase)	Kimura et al[108]
Male Wistar rats ALDO mice (ALDO (0.5 μg/h) for 4 wk).	Dosage: SPL (100 mg/kg per day) mixed in food Duration: 4 wk	↓Atrial tissue remodeling (↓fibrosis) ↓Atrial arrhythmia (↓AF duration, ↑APD) ↓Atrial molecular remodeling (↓Kir2.1 and Kv1.5)	Lammers et al[98]
MHC-TGF-β1cys33ser transgenic mice (develop selective atrial fibrosis at the age of 3.5-mo-old)	Dosage: EPL (200 mg/kg per day) with food Duration: 75 d	↓Atrial tissue remodeling (↓fibrosis) ↓Atrial molecular remodeling (↓macrophage infiltration, ↓CTGF, ↓fibronectin)	Chen et al[109]
MHC-TGF-β1cys33ser transgenic mice (develop selective atrial fibrosis [3.5 mo of age])	Dosage: EPL (200 mg/kg per day) with food Duration: from age 28 d to 105 d (≈11 wk)	↓Atrial tissue remodeling (↓fibrosis) ↓Atrial molecular remodeling (↓fibronectin, ↓CTGF, ↑Smad2/3, and Smad7).	Du et al[110]
Male sheep (=34) Model: atrial tachypacing-induced persistent AF (tachypacing: 13 Hz (780 bpm) for up to 23 wk) Duration: 108.4±11.4 d (SP animals), 125.3±10.9 d (EPL animals), and 193±45.5 d in sham animals	EPL (100 mg/d), orally.Duration: from the start of tachypacing until end point	↓Atrial remodeling (↓LA area) ↓Atrial arrhythmia (↓AF wave complexity, ↓AF burden, and inducibility)↓atrial tissue remodeling (↓cardiomyocyte size, ↓fibrosis) ↓Atrial molecular remodeling (↓SMA, ↓COLIII)	Takemoto et al[111]
Male Sprague-Dawley rats CIH to simulate obstructive sleep apnea. 6 wk of daily CIH treatment.	EPL (10 mg/kg per day), oral gavage Duration: 6 wk (concurrent with CIH treatment)	↓Atrial remodeling (↓LA diameter) ↓Atrial arrhythmia (↓AF inducibility) ↓Atrial tissue remodeling (↓fibrosis) ↓Atrial molecular remodeling (↓JAK/STAT3) ↓Atrial electrical remodeling (↑ERP).	Yang et al[112]

AF indicates atrial fibrillation; ALDO, aldosterone; APD, action potential duration; Bax, Bcl-2-associated X protein; Bcl-2, B-cell lymphoma 2; CCL2, C-C motif chemokine ligand 2; CIH, chronic intermittent hypoxia; COLIII, type III collagen; CTGF, connective tissue growth factor; DS, Dahl salt-sensitive; EPL, eplerenone; ERP, effective refractory period; HNE, 4-hydroxynonenal; HTN, hypertension; $I_{Ca,L}$, L-type calcium current; JAK/STAT3, Janus kinase/signal transducer and activator of transcription 3; Kir2.1, inwardly rectifying potassium channel; Kv1.5, voltage-gated potassium channel 1.5; LA, left atrium; LAAEF, left atrial appendage ejection fraction; LAAVmax, maximum left atrial appendage volume; LAAVmin, minimum left atrial appendage volume; LAD, left anterior descending artery; LAEF, left atrial ejection fraction; LAVmax, maximum left atrial volume; LAVmin, minimum left atrial volume; MI, myocardial infarction; MMP-9, matrix metallopeptidase 9; MRA, mineralocorticoid receptor antagonist; OPN, osteopontin; P-wave, the part of the ECG that reflects atrial depolarization; Rac1, Ras-related C3 botulinum toxin substrate 1; SMA, smooth muscle actin; Smad2/3, mothers against decapentaplegic homolog 2/3; Smad7, mothers against decapentaplegic homolog 7; SPL, spironolactone; TGF-β1, transforming growth factor beta 1; TIMP-1, tissue inhibitor of metalloproteinases 1; TNF-α, tumor necrosis factor-alpha.

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signaling downstream of the TGFβ receptor. This may include inter-organ crosstalk, since in the same mouse model, MR deletion from osteoblasts prevented atrial fibrosis.[119] Osteoblasts secrete osteocalcin to the circulation, which triggers the activation of GPRC6A receptors on atrial fibroblasts, downstream PKA (protein kinase A)/CREBH (cAMP-responsive element-binding protein H) signaling, and fibroblast proliferation and migration, which can be inhibited by MR deletion.[119]

Taken together, MR activation in atrial fibroblasts results in profibrotic gene expression, contributing to atrial tissue fibrosis and inflammation. MR activation in extracardiac cells like osteoblasts indirectly effects on atrial fibroblasts (Figure 1). However, the mechanisms

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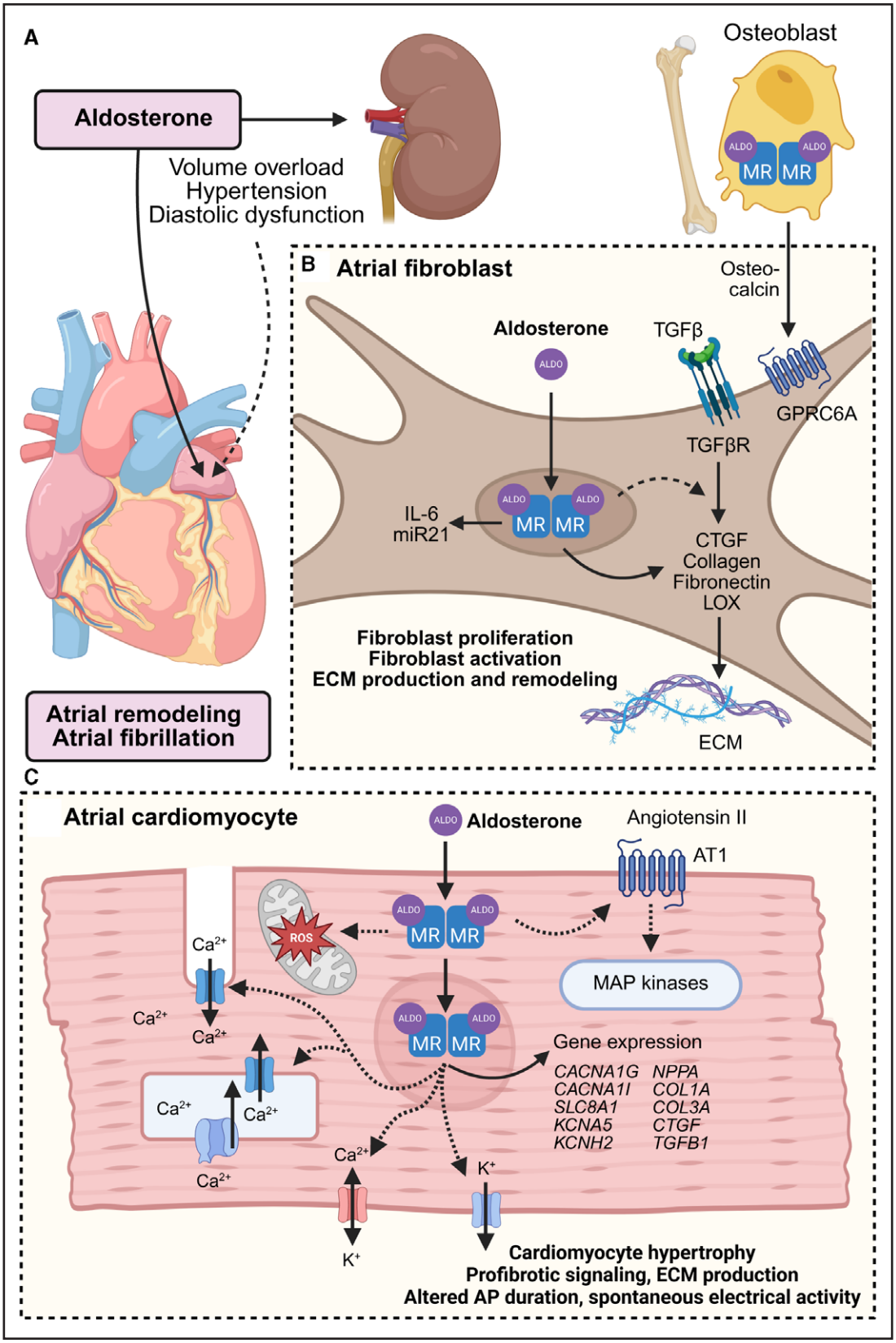

Figure 1. Impact of the MR (mineralocorticoid receptor) in atrial cardiac fibroblasts and cardiomyocytes on atrial remodeling. **A**, Aldosterone via direct action on atrial cells and indirectly by causing volume overload, hypertension, and diastolic dysfunction drives atrial remodeling and increases the risk for atrial fibrillation. **B**, In cardiac fibroblasts, aldosterone-mediated MR activation modulates regulation of the ECM (extracellular matrix), including the production of ECM components, remodeling enzymes, and profibrotic and proinflammatory regulatory molecules. MR may interfere with TGFβ (transforming growth factor beta) signaling, which is also responsible for (*Continued*)

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by which MR interacts with other signaling pathways, for example, potential direct interplay between MR and TGFβ signaling, in promoting atrial fibrosis require further investigation.

MR SIGNALING IN ATRIAL CARDIOMYOCYTES

Cardiomyocyte contraction requires a tight regulation of intracellular calcium flux that is achieved via channels and ATP-dependent calcium pumps. In AF, cardiomyocytes undergo structural remodeling and show disturbed calcium handling, leading to altered AP formation and propagation and impaired contractile function.[8] MR is expressed in cardiomyocytes and contributes to adverse remodeling and left ventricular dysfunction after myocardial infarction, pressure overload, or doxorubicin treatment.[26,27,120] Several findings point towards enhanced MR signaling in cardiomyocytes during AF: atrial cardiomyocytes show high MR expression[103] that can be further upregulated by electrical activity as observed in AF. Rapid electrical stimulation of atrial cardiomyocytes increases MR expression through a calcium-dependent mechanism.[91] Similarly, arrhythmic pacing of neonatal rat cardiomyocytes increases 11β-HSD2 expression,[94] thereby enhancing MR signaling. Aldosterone may induce a feed-forward cycle by raising MR and AT1 (angiotensin II type 1 receptor) expression in atrial cardiomyocytes by activating various members of the MAPK (mitogen-activated protein kinase) signaling pathway family, including ERK (extracellular signal–regulated kinase), JNK (c-Jun N-terminal kinase), and p38 MAPK pathways.[121] MR activation in atrial cardiomyocytes is associated with increased expression of ECM components, including COL1A and COL3A (collagen 1A and 3A), TGF-β1, and α-SMA.[121] In mice, aldosterone treatment leads to a shift in cardiomyocyte subpopulations and the dysregulation of genes related to calcium handling and fibroblast activation in atrial cardiomyocytes[103] (Figure 2A). Interestingly, a recent study has revealed that patients with AF have an increased subset of atrial cardiomyocytes with elevated MR expression, associated with inflammation and stress-related genes.[65]

MR activation promotes ROS production in cardiomyocytes via genomic and nongenomic effects.[26,91,122,123] MR-mediated stimulation of NOX enzyme family members, including NOX1, NOX2, and NOX4 modulates ROS production in cardiomyocytes.[123] The resulting accumulation of ROS and attenuated NO production contributes to aldosterone-induced atrial cardiomyocyte hypertrophy.[122] In addition, excessive ROS in cardiomyocytes activates Src, Smad (mothers against decapentaplegic homolog) 2/3, and ERK1/2 signaling pathways, upregulating voltage-gated potassium channel 1.5 (Kv1.5).[123] The Kv1.5 potassium channel contributes to the ultra-rapid rectifier potassium current (I_{Kur}) in atrial cardiomyocytes, essential for the AP repolarization, affecting AP duration.[124] Enhanced Kv1.5 expression or activity shortens AP duration, which is a characteristic of atrial proarrhythmic electrical remodeling.[125]

MR SIGNALING IN ELECTRICAL REMODELING

Several studies have demonstrated that the MR signaling in atrial cardiomyocytes modulates activity and expression of ion channels that are crucial for atrial cardiomyocyte electrophysiology, influencing AP duration, repolarization, and overall cardiac rhythm (Figure 2B). For instance, aldosterone increases L-type calcium current in atrial cardiomyocytes, resulting in an increase in calcium transient amplitude, prolongation of AP duration, a higher incidence of unsynchronized calcium oscillations, and an increase in cell membrane capacitance.[126] This L-type calcium current enhancement occurs through nongenomic effects of aldosterone involving cAMP/PKA signaling–dependent phosphorylation of both CaV1.2 and CaV1.3 L-type calcium channels.[126] Similarly, aldosterone stimulation also increases T-type calcium current activity, attributed to MR-mediated increased expressions of α-1G (CaV3.1) and α-1H subunits (CaV3.2).[91] In turn, increased T-type calcium current activity contributes to early depolarization and may enhance arrhythmogenicity via AD duration shortening.[127] Furthermore, aldosterone augments sarcoplasmic reticulum calcium content and calcium transient magnitude, leading to atrial cardiomyocyte hypertrophy via MR signaling.[91,122] In addition, aldosterone upregulates NCX (Na^+-Ca^{2+} exchanger) expression in atrial cardiomyocytes via MR signaling.[122] NCX mediates calcium efflux during diastole, although its precise role in MR signaling effects on atrial cardiomyocytes remains unclear.

Potassium channels are also crucial in modulating the electrical activity of atrial cardiomyocytes. MR activation

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Figure 1 Continued. the regulation of several aspects of ECM regulation. In addition, activation of the MR in bone osteoblasts increases the synthesis and secretion of osteocalcin, which in turn activates the GPRC6A (G-protein–coupled receptor family C group 6 subtype A) receptor on atrial fibroblasts, altogether driving fibroblast proliferation, activation, and ECM remodeling. **C**, In cardiomyocytes, genomic MR effects augment the expression of genes related to cardiomyocyte hypertrophy, fibroblast activation, and ion channel activity. MR via nongenomic signaling activates nicotinamide adenine dinucleotide phosphate (NADPH) oxidases, leading to an increase in reactive oxygen species (ROS) production, interacts with AT1 (angiotensin II type 1 receptor), modulating MAP (mitogen-activated protein) kinases, and changes ion channel activity. AP indicates action potential; CTGF, connective tissue growth factor; IL, interleukin; LOX, lysyl oxidase; miR21, microRNA-21; and TGFβR, transforming growth factor β receptor.

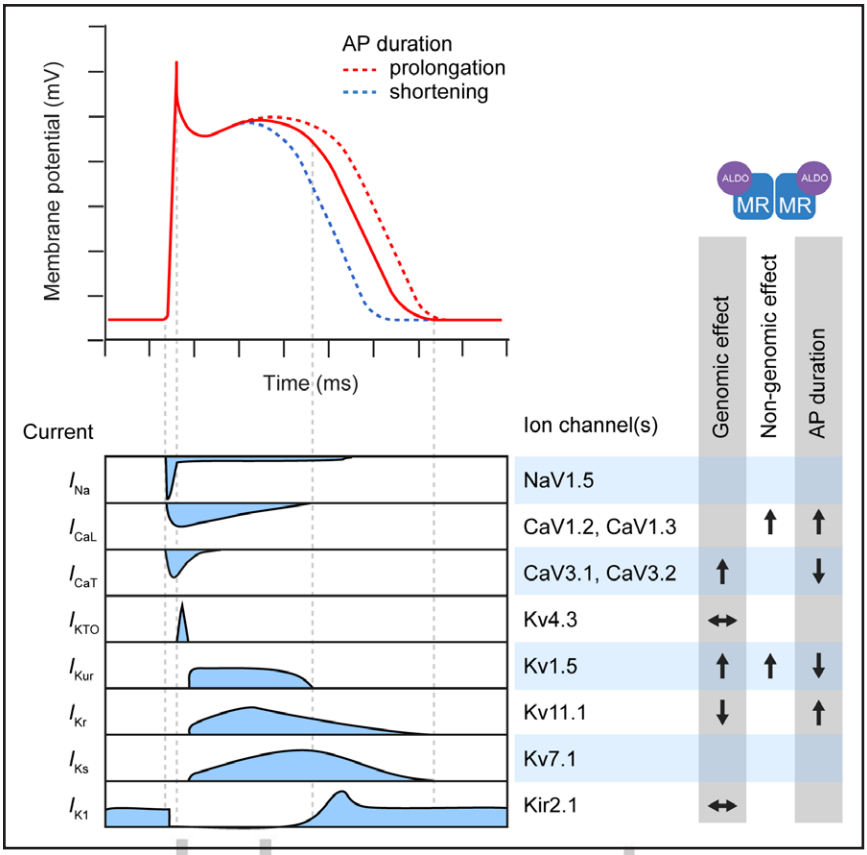

Figure 2. Regulation of the cardiomyocyte action potential by MR (mineralocorticoid receptor) signaling.
MR signaling can impact various ion channels, which in turn can alter the duration and morphology of the action potential (AP) in atrial cardiomyocytes. The AP duration can either be prolonged or shortened due to MR-mediated alterations of ion channel expressions or activity via genomic and nongenomic MR action. I_{CaL} indicates L-type calcium current; I_{CaT}, T-type calcium current; I_{K1}, inward rectifying potassium current; I_{Kr}, rapid delayed rectifier potassium current; I_{Kto}, transient outward potassium current; and I_{Kur}, ultra-rapid rectifier potassium current.

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reduces rapid delayed rectifier potassium current density in atrial cardiomyocytes, prolonging AP duration through decreased *ERG* subunit expression (KCNH2, Kv11.1).[91] Conversely, aldosterone treatment increases I_{Kur} without significantly influencing other potassium currents, like the inward rectifying potassium current, the transient outward potassium current.[126] Although a responsible potassium channel was not identified,[126] evidence suggests that Kv1.5 channels mainly contribute to I_{Kur}.[128,129] Moreover, aldosterone acts via the MR to activate Nox/ROS signaling, stimulating Smad2/3, ERK1/2, and Src signaling pathways, increasing Kv1.5 channel expression in atrial cardiomyocytes,[123] responsible for I_{Kur} current.[124] Furthermore, aldosterone facilitates the redistribution of Kv1.5 channels from the intercalated disc region to the lateral plasma membrane and Z-disc/T-tubulus region through specific Rab GTPases (Rab4, Rab5, and Rab9) expression, activating the Akt/AS160 pathway.[130] A study using human induced pluripotent stem cell–derived cardiomyocytes with atrial cardiomyocyte phenotypes showed that increasing I_{Kur} density shortens AP duration.[131] However, specific I_{Kur} blockers demonstrated I_{Kur} activity effects on AP duration differ between homeostatic and pathological states.[132] Other ion channels that may be of relevance in AF, such as TASK-1 or Kv1.2, have—to our knowledge—not been assessed in the context of MR. In conclusion, MR plays a significant role in the regulation of ion channels and currents in atrial cardiomyocytes, influencing their expression, activity, and distribution within the cell (Figure 2B).

SUMMARY

The comprehensive evidence from clinical trials, experimental models, and cellular mechanistic studies summarized here demonstrates that MR activation drives both structural and electrical atrial remodeling processes and thereby contributes to AF development and recurrence. Clinical studies indicate potential therapeutic benefits of MRA across diverse patient populations, ranging from patients with HF to those with hypertension and primary aldosteronism, to patients undergoing cardiac surgery, with MRAs improving atrial remodeling and reducing AF burden (Table 1). Notably, recent trials using finerenone in patients with diabetes and chronic kidney disease support the concept of a preventive rather than exclusively reactive approach to AF in high-risk patient collectives. This preventive effect aligns with the pathophysiological rationale that aldosterone-MR signaling contributes to the formation of an arrhythmogenic substrate rather than solely sustaining established electrical instability.

Preclinical studies demonstrate that aldosterone infusion models consistently induce atrial remodeling when combined with additional stressors, supporting a multihit hypothesis for AF development (Table 3). The

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protective effects of MRA exhibit consistent reduction in atrial dilatation, fibrosis, and electrical remodeling across various animal models (Table 4). The cellular mechanisms underlying these effects reveal the complex interplay between MR signaling in different atrial cell types. In atrial fibroblasts, MR activation modulates diverse proinflammatory and profibrotic factors, thereby promoting atrial tissue fibrosis. In atrial cardiomyocytes, MR signaling profoundly affects ion channel by modulating calcium and potassium channel expression, consequently predisposing atrial tissue to arrhythmogenic activity generation. Selective MR knockout models could help to further separate the role of different cell types and provide insights into the molecular mechanisms involved. Particularly whether MR signaling in endothelial cells, smooth muscle cells, and inflammatory cells within atrial tissue contributes to adverse atrial remodeling and arrhythmia warrants future investigation.

FUTURE PERSPECTIVES AND CLINICAL IMPLICATIONS

The accumulating evidence supporting the role of MR signaling in AF pathogenesis has significant implications for clinical practice and future research directions. The consistently beneficial effects of MRA across diverse patient populations suggest a broader role in AF prevention and management than currently recognized and support further investigation. First, large-scale clinical trials are needed to demonstrate that MRA use can improve relevant clinical end points in patients at risk or with preexisting AF. Mechanistically, the bidirectional relationship between aldosterone levels and AF progression raises questions regarding causality, necessitating longitudinal investigation. Existing evidence from experimental studies indicates that aldosterone may function as an initiating factor of atrial remodeling, whereas AF may serve as a stimulus for subsequent aldosterone elevation. In addition, aldosterone may cause hemodynamic effects that increase mechanical stress to atrial tissue and thereby further exacerbate remodeling. These interactions establish a potential positive feedback loop, which could contribute to the progressive nature and therapeutic resistance of AF. Although there is convincing evidence that aldosterone causes structural remodeling of atrial tissue, this has not been specifically assessed for pulmonary veins, which should be addressed in future studies.

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Disclosures

None.

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